Investigating AI Adoption in SMEs

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Abstract

As small to medium-sized enterprises (SMEs) increasingly integrate generative artificial intelligence (AI) to enhance productivity and maintain competitiveness, understanding employee readiness to adopt and effectively use these technologies becomes important. This policy paper explores the views and concerns of SME employees towards generative AI, focusing on their willingness to learn and the factors influencing their adoption. Using a mixed-methods approach, this study combines structured interviews and a quantitative survey to provide a broad view of the current knowledge levels, concerns and preferred learning methods of employees.

The results show that employee readiness to use AI is shaped by their knowledge of AI and concerns about data privacy and job security, with age not being a significant factor. Furthermore, despite widespread concerns about AI, there is a significant willingness among employees to engage with AI tools to improve efficiency and productivity.

This paper suggests that by addressing the identified barriers and leveraging preferred learning methods, SMEs can develop a workforce skilled in AI. Recommendations are provided to support SMEs in crafting strategies that not only mitigate fears associated with AI but also enhance the workforce's capabilities to utilize the potential of generative AI effectively, thereby promoting a more efficient adoption and utilization of these technologies within small to medium-sized enterprises.

1 Introduction

As AI, particularly generative AI, continues to transform industries, businesses are seeking ways to integrate AI into their operations to enhance productivity and remain competitive. Generative AI offers powerful tools for increasing productivity, creativity, and efficiency across diverse roles. This leads to SMEs facing unique challenges adopting generative AI, but also ensuring that their workforce is sufficiently equipped to leverage these technologies effectively. However, as SMEs explore these technologies, their employees' readiness to engage with AI can be influenced by a range of factors, including age, existing knowledge, and personal concerns about AI.

The research investigates four questions related to the attitudes and concerns of SME employees about the adoption of (generative) AI.

Firstly, the relation of SME employees' negative concerns about AI and their willingness to learn about AI is investigated, recognizing that concerns around job security, data privacy, and ethical issues may play a role in the willingness to learn more about AI.

Second, the correlation between SME employees' age and their likelihood of embracing generative AI in the workplace activities is researched, considering generational differences in technology familiarity and openness to change.

Third, the research identifies key factors contributing to SME employee concerns and misconceptions about AI, shedding light on how misperceptions may hinder AI adoption in organizations.

Finally, the influence of existing generative AI knowledge among SME employees on an employee's enthusiasm to pursue further learning of generative AI is investigated.

By understanding these dynamics, a supportive environment for AI adoption that addresses employee concerns, dispels misconceptions, and promotes learning can be supported. Through this analysis, the paper provides insights to guide policies and strategies aimed at bridging knowledge gaps, building trust and enhancing AI readiness within SMEs, ultimately enabling SMEs to create effective strategies to capitalize on the potential of AI.

2 Approaches and Results of Research

The research adopted a mixed-methods approach, integrating both qualitative and quantitative methods to provide a whole understanding of employee attitudes towards AI. The qualitative part of this study consists of structured interviews, while the quantitative aspect consists of surveys distributed through Prolific. To test the relationship between age and willingness to adopt generative AI tools, a two-sample t-test (Welch's t-test) was performed. For further details and examples, refer to the book OpenIntro Statistics [1].

2.1 Qualitative Data Collection

In the qualitative studies, structured interviews were conducted by each team member with six participants from various SMEs, ensuring a range of perspectives based on industry and job function. The interviews were designed to explore participants' understanding of AI, their concerns, and their willingness to learn about AI applications. Each interview lasted approximately 20-30 minutes and was recorded with participants' consent. An interview guide was created in advance, containing 13 core questions aimed at inducing rich, descriptive responses.

Before the interviews, participants received an information letter outlining the study's purpose and their rights as participants, followed by an informed consent form. The interviews were conducted both in-person and online, utilizing recording tools suitable for each format. The collected qualitative data was transcribed, encoded, and analyzed thematically to identify key patterns and insights.

2.2 Quantitative Data Collection

In the quantitative study, a structured survey was distributed through Prolific, a platform known for connecting researchers with a broad demographic. The survey assessed employees' current knowledge of AI, their willingness to adopt generative AI tools, and the contextual factors influencing AI usage in the workplace. The survey included sections on demographics, current AI knowledge, willingness to adopt AI, and learning preferences, allowing for a systematic exploration of the participants' backgrounds and attitudes.

The survey questions were carefully created in order to capture a wide range of responses, including willingness to dedicate time to AI learning and interest in various AI topics. Data collected from the surveys provided quantitative support for the qualitative findings, enhancing the overall understanding of employee attitudes toward AI.

2.3 Data Analysis

For the qualitative data analysis, transcriptions of the interviews were subjected to thematic analysis. Open coding was applied to identify initial themes, which were then refined through axial coding to establish broader categories. Since the analysis focused from the beginning on understanding how concerns impact learning, selective coding was naturally integrated. Major themes included "Job Loss Concerns," "Positive Attitudes Toward Learning AI," and "Awareness and Understanding of AI.". The qualitative insights were complemented by the quantitative data, which showed that employees with a greater understanding of generative AI are more inclined to engage in additional learning opportunities related to these technologies.

2.4 Results

In this section, the results of the study will be discussed in depth. Demographic profiles and descriptive findings are part of the analysis.

2.4.1 Demographic profiles of the respondents

As previously stated, there were a total of 160 valid responses. Appendix II provides an overview of the demographics of the sample used in this research. As shown in the table, most respondents are from Africa (~53%), followed by Europe (~38%). The majority of the respondents identify as female (~65%), while a smaller percentage identify as male (~34%). In terms of education, most respondents hold a bachelor's degree (~54%), with fewer holding a master's degree (~23%) or a high school diploma (20%),

Regarding organizational size, the largest group of respondents work in medium-sized organizations (~43%), followed by small (~34%) and micro (~19%) organizations.

In terms of age, the sample predominantly comprises individuals aged 25-34 (~43%) and 18-24 (~29%), while older age groups are underrepresented: only ~17% are aged 35-44, ~9% are aged 45-54, and a mere ~1% are aged 55-64.

2.4.2 Descriptive results, AI knowledge and willingness to learn

Table 1 shows the overall averages and standard deviations for all the variables included in the survey. Results were normalized on a scale from 0 to 1, in which the means will range. The neutral response 0.5 (neither agree nor disagree).

This means that a mean under 0.5 is perceived as positive, while a mean below 0.5 is perceived as negative. Table 2 shows that, on average, respondents show

Table 1: Descriptive statistics for variables

Variables	**Mean**	**Standard Deviation**
Current knowledge of AI	0.511	0.152
Interest in learning new AI tools	0.847	0.214
Time to spend on learning new AI tools	0.434	0.231
Knowledge Metric	0.522	0.212
Willingness Metric	0.640	0.186

an average level of self-perceived generative AI knowledge (M≈0.511). This is also visible in the knowledge metric (M≈0.522). Respondents do show to have a high level of interest in learning about new AI tools (M≈0.847). Again, this can also be found in the willingness metric (M≈0.640).

Qualitative responses highlighted several barriers to adopting generative AI technologies. Common concerns included a lack of time, inadequate resources, and fears of job displacement. These insights indicate the importance of addressing employees' apprehensions and providing support to facilitate successful adoption.

2.4.3 Existing AI knowledge and willingness to learn

The objective is to establish whether higher generative AI knowledge leads to more willingness to learn about generative AI. The hypotheses can be formulated statistically after calculating the critical value for a two-tailed r-test using $df = 158$ and $\alpha = 0.05$ The following statistical hypotheses can be formulated after discovering the values:
$H_0 : r \leq 0.1552$
$H_A : r > 0.1552$
This leads to the State Decision Rule: if r is greater than 0.1552, reject the null hypothesis. The Pearson Correlation Coefficient [1] is calculated using Python as the statistical software. This indicates that the Pearson's r = 0.402, thus the null hypothesis can be rejected. State conclusion: there is a relationship between a SME employee's existing generative AI knowledge and their willingness to learn more about generative AI. $r(158) = 0.402, p < 0.05$.

2.4.4 Existing AI knowledge and age

The quantitative analysis indicated that the average willingness score for younger employees was 7.2 out of 10, while older employees averaged 6.8. Although younger employees displayed slightly higher enthusiasm for generative AI adoption, the two-sample t-test produced a p-value of 0.374, indicating no statistically significant difference between the two age groups. Thus, the null hypothesis could not be rejected, suggesting that age may not be a decisive factor influencing the willingness to adopt generative AI. Both age groups reported comparable levels of knowledge regarding generative AI technologies. Approximately 65% of respondents identified as having moderate to high levels of understanding. This finding suggests that educational initiatives can effectively reach employees of all ages, emphasizing the need for continuous learning opportunities.

3 Conclusions

This study aimed to understand what the current level of knowledge about generative AI among employees in SMEs is, how willing they are to adopt it in their roles, and what is their preferred way of acquiring these skills. After conducting the survey and the interviews, this research found that AI is growing throughout all company sizes, especially SMEs. The adoption of AI technology within SMEs becomes more prominent nowadays, with the rise of generative AI helping employees automate time consuming tasks. Employees with a greater understanding of generative AI show higher interest in learning about these technologies.

4 Recommendations

This study has revealed key insights into the implementation of generative AI into SMEs, going over current knowledge of employees, their willingness to adopt and their preferred methods of learning. The findings highlighted improvements that can be implemented in SMEs.

Our research revealed that having prior knowledge of generative AI and AI in general helped strengthen employees' willingness to learn further. It is recommended that employees looking to learn about generative AI are exposed to a basic tutorial or course covering the learning base needed to further learn about the AI tool. As in our research, employees feel too overwhelmed when starting out and end up never learning or using generative AI due to there being a learning barrier there. Explaining what generative AI is, what it can do and how to effectively use it in basic terms would help keep employees interested and open to more advanced concepts and techniques involving generative AI.

As shown in the research, currently employees use a more self-paced approach to learning about generative AI. Some of the popular methods gathered and those that are recommended for employees include YouTube videos and online courses. There are many helpful and easy to understand online YouTube sources on generative AI. Using generative AI itself is an excellent way of learning about the topic and builds experience using generative AI. Sources such as ChatGPT and Copilot work well. Literature and AI books are also a good way to learn about the topic if reading books is the preferred method of learning.

Another thing that was discovered is there was not enough support from the management for employees to learn about generative AI in the workplace. It is recommended that the management of companies evaluate what they offer in terms of training opportunities and consider how to encourage employees to participate in more learning opportunities.

References

[1] David M. Diez, Christopher D. Barr, and Mine Çetinkaya-Rundel. *OpenIntro Statistics*. Fourth Edition, OpenIntro, Inc., Boston, MA, 2019. ISBN: 978-1-7327752-0-3. Available at: `https://www.openintro.org/book/os/`.

5 Appendix

5.1 Appendix I: Stakeholder analysis

1. SME Employees

   Why They Matter: These are the primary subjects of our research. Their knowledge levels, perceptions, concerns, and willingness to learn about AI directly impact the adoption and integration of AI in SMEs.

   Concerns: Job security, relevance of their skills in an AI-driven workplace, and opportunities for upskilling.

2. SME Owners and Managers

   Why They Matter: These decision-makers are responsible for the strategic direction of their businesses, including the adoption of new technologies like AI. They need to understand the implications of AI on their workforce and how best to implement it.

   Concerns: Return on investment in AI, employee retention and morale, competitive advantage, and efficient resource allocation for training.

3. Industry Associations and Chambers of Commerce

   Why They Matter: These organizations represent the collective interests of SMEs and can influence policy, provide training, and support the adoption of AI across the sector. So basically, Digi.

   Concerns: Supporting member businesses in navigating AI adoption, advocating for SME-friendly policies, and providing resources for upskilling.

4. Government and Policymakers

   Why They Matter: Policymakers at the local, regional, and national levels can influence the regulatory environment, provide funding for training programs, and develop policies that support SME growth and technology adoption.

   Concerns: Economic growth, workforce development, job creation, and ensuring equitable access to AI technologies across different sectors and demographics.

5. Educational Institutions and Training Providers

   Why They Matter: These stakeholders are responsible for designing and delivering training programs that can help SME employees acquire the necessary skills to work with AI.

   Concerns: Developing relevant curricula, bridging the skill gap, and providing accessible education for upskilling and reskilling.

6. AI and Technology Developers

   Why They Matter: Companies that develop and sell AI technologies are interested in the successful adoption of their products by SMEs. They play a role in educating potential customers and ensuring their products are user-friendly and beneficial.

   Concerns: Market adoption, customer satisfaction, and developing AI solutions that address the specific needs of SMEs.

7. Customers and Clients of SMEs

   Why They Matter: Although indirectly impacted, the adoption of AI by SMEs can affect the quality, efficiency, and type of services/products they receive. Customer perceptions can influence the success of AI implementations.

   Concerns: Service quality, responsiveness of SMEs to market changes, and ethical considerations in AI usage (e.g., data privacy).

5.2 Appendix II: Demographic Results

Table 2: Demographic Characteristics of the Sample (N=160)

Demographic Characteristics	**Frequency (n)**	**Percentage (%)**
Geographic		
Africa	84	52.500
Europe	61	38.125
North America	11	6.875
Asia	3	1.875
Oceania	1	0.625
Gender		
Female	104	65.00
Male	54	33.75
Non-binary / third gender	2	1.25
Highest education		
Bachelor's degree	87	54.375
Master's degree	37	23.125
High school diploma or equivalent	32	20.000
Doctoral degree	3	1.875
Less than high school	1	0.625
Organizational size		
Micro (1-9 employees)	31	19.375
Small (10-49 employees)	55	34.375
Medium (50-249 employees)	69	43.125
Do not know	5	3.125
Employee age		
18-24	47	29.375
25-34	69	43.125
35-44	27	16.875
45-54	15	9.375
55-64	2	1.250